Spring, 2017

STAT 2056 Probability and Random Variables
Midterm Exam - Solutions

1. E = "error", E_1 = "type 1 error", E_2 = "type 2 error"

(a) $P[E=4] = \binom{10}{4}(0.001)^4(0.999)^6$ (10 pts)

(b) "Type 1 error" occurs with probability $0.4 \times 0.001 = 0.0004$. (5 pts) Hence $P[E_1=3] = \binom{10}{3}(0.0004)^3(0.9996)^7$ (10 pts)

2. (a) Series system: $A_1 \cap A_2 \cap A_3$ (2), Parallel system: $A_1 \cup A_2 \cup A_3$ (4)

Two-out-of-three system $(A_1 \cap A_2 \cap A_3') \cup (A_1^c \cap A_2 \cap A_3) \cup (A_1 \cap A_2^c \cap A_3) \cup (A_1 \cap A_2 \cap A_3^c)$ (4)

(b) Series system: $P(A_1 \cap A_2 \cap A_3) = P(A_1).P(A_2).P(A_3) = 0.8 \times 0.75 \times 0.9 = 0.54$ (5)

Parallel system: $P[A_1 \cup A_2 \cup A_3) = P[A_1]+P[A_2]+P[A_3] - \{P(A_1 \cap A_2)+P(A_1 \cap A_3)+P(A_2 \cap A_3)\} + P(A_1 \cap A_2 \cap A_3)$

$= 0.8+0.75+0.9 - \{0.8\times0.75+0.8\times0.9+0.75\times0.9\} + 0.8\times0.75\times0.9$

$= 2.45 - 1.995 + 0.54 = 0.995$ (5)

Two-out-of-three system: $P(A_1 \cap A_2 \cap A_3) + P(A_1^c \cap A_2 \cap A_3) + P(A_1 \cap A_2^c \cap A_3) + P(A_1 \cap A_2 \cap A_3^c)$

$= 0.8\times0.75\times0.9 + 0.2\times0.75\times0.9 + 0.8\times0.25\times0.9 + 0.8\times0.75\times0.1$

$= 0.54 + 0.135 + 0.18 + 0.06 = 0.915$

3. (a)

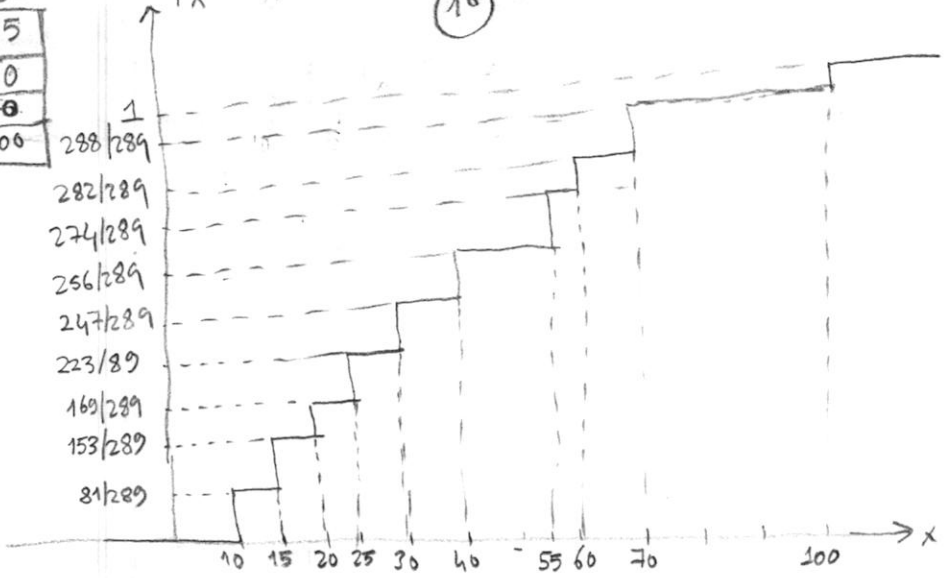
First outcome \ second outcome	5	10	20	50
5	10	15	25	55
10	15	20	30	60
20	25	30	40	~~70~~
50	55	60	70	100

(5)

Sum=k	P[X=k]
10	$\frac{9}{17}\cdot\frac{9}{17} = \frac{81}{289}$
15	$2\cdot\frac{9}{17}\cdot\frac{4}{17} = \frac{72}{289}$
20	$\frac{4}{17}\cdot\frac{4}{17} = \frac{16}{289}$
25	$2\cdot\frac{9}{17}\cdot\frac{3}{17} = \frac{54}{289}$
30	$2\cdot\frac{4}{17}\cdot\frac{3}{17} = \frac{24}{289}$
40	$\frac{3}{17}\cdot\frac{3}{17} = \frac{9}{289}$
55	$2\cdot\frac{9}{17}\cdot\frac{1}{17} = \frac{18}{289}$
60	$2\cdot\frac{4}{17}\cdot\frac{1}{17} = \frac{8}{289}$
70	$2\cdot\frac{3}{17}\cdot\frac{1}{17} = \frac{6}{289}$
100	$\frac{1}{17}\cdot\frac{1}{17} = \frac{1}{289}$

(10)

(b) $F_X[x] = P[X \le x]$ (10)



4. (a)

X	Y	P[Y]
-3	1	1/10
-2	4	2/10
-1	7	3/10
0	10	4/10

(5)

$\phi_y(w) = \sum_y P[y]\, e^{jwy}$

$= \frac{1}{10}\cdot e^{jw1} + \frac{2}{10}e^{jw4} + \frac{3}{10}e^{jw7} + \frac{4}{10}e^{jw10}$

$= \frac{1}{10}\left[e^{jw} + 2e^{j4w} + 3e^{j7w} + 4e^{j10w}\right]$ (5)

(b) $\frac{d\phi_y(w)}{dw} = \frac{j}{10}\left[e^{jw} + 8e^{j4w} + 21e^{j7w} + 40e^{j10w}\right]$ (3)

$\frac{d^2\phi_y(w)}{dw^2} = \frac{j^2}{10}\left[e^{jw} + 32e^{j4w} + 147e^{j7w} + 400e^{j10w}\right]$ (3)

$E[Y] = \frac{1}{j}\left.\frac{d\phi_y(w)}{dw}\right|_{w=0} = \frac{1}{10}[1+8+21+40] = 7$ (3)

$E[Y^2] = \frac{1}{j^2}\left.\frac{d^2\phi_y(w)}{dw^2}\right|_{w=0} = \frac{1}{10}[1+32+147+400] = 58$ (3)

$Var(Y) = E[Y^2] - (E[Y])^2 = 58 - 7^2 = 9.$ (3)